

BE 201: Cell Biology

Course Code: BE 201

Course Name: Cell Biology

L-T-P-C: 3-0-2-4

Prerequisite: None

Students intended for: B. Tech M.Tech Integrated Dual Degree Bioengineering students

Approval: 28th Senate; 44 BoA

Course contents

- **Cells: The fundamental units of life**

The structure and ultrastructure of the cell (the prokaryotic cell and the eukaryotic cell). Membrane enclosed Organelles (nucleus, chloroplast, mitochondria, ribosome etc) Model organisms (Yeast; simple eukaryotic cell, Arabidopsis; model plant, Fish/worms/mice; model animal [8 Lectures]

- **Cell Membranes and membrane transport**

Membrane structure (lipid bilayer, and membrane protein), Passive and Active Transport, Ion channels, ATP pumps. Na^+ / K^+ / Ca^{2+} pumps uniport, symport antiporter system. Ligand gated / voltage gated channels, Agonists and Antagonists [8 Lectures]

- **The cell-division cycle**

Cell cycle – Mitosis, Meiosis, Molecules controlling cell cycle, Extra cellular matrix, role of matrix in cell adhesion: Gap junctions, Tight junctions, Desmosomes, Hemidesmosomes [8 Lectures]

- **Cell-cell communication and protein transport**

General principles of cell signaling, G-protein coupled receptor, enzymes-coupled receptor, protein sorting, vesicular transport, endocytic pathways [8 Lectures]

- **Techniques used to study cells**

Cell fractionation, Morphology and identification of cells using microscopic studies like SEM, TEM and Confocal Microscopy. Localization of proteins in cells – Immunostaining. [10 Lectures]

Lab

List of Experiments

- Introduction to principles of sterile techniques and cell propagation
- Principles of microscopy, phase contrast and fluorescent microscopy
- Gram's Staining
- Leishman Staining
- Trypan Blue Assay
- Staining for different stages of mitosis in Allium Cepa (Onion)



Text Books:

1. Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, **Molecular Biology of the Cell**, 6th Edition, Garland Science, 2014.

References:

1. Harvey Lodish, Arnold Berk, Chris A. Kaiser, Monty Krieger, Anthony Bretscher, Hidde Ploegh, Angelika Amon, Kelsey C. Martin, **Molecular Cell Biology** - 8th Edition, W.H. Freeman, 2016.
2. George Plopper, David Sharp, Eric Sikorski, **Lewin's Cells**, 3rd Edition, Johns & Bartlett Publishers, 2015
3. Lisa A. Urry, Michael L. Cain, Steven A. Wasserman, Peter V. Minorsky, Jane B. Reece, **Campbell Biology**, 11th Edition, Pearson, 2020.